Neural Speech Synthesis: A Review

Speech synthesis is the artificial production of human speech. A computer system used for this purpose is called a speech synthesizer, and it can be implemented in software or hardware products. A text-to-speech system converts normal language text into speech; other systems render symbolic linguistic representations like phonetic transcriptions into speech.

Synthesized speech can be created by concatenating pieces of recorded speech that are stored in a database. Systems differ in the size of the stored speech units; a system that stores phones or diphones provides the largest output range, but may lack clarity.

For specific usage domains, the storage of entire words or sentences allows for high-quality output. Alternatively, a synthesizer can incorporate a model of the vocal tract and other human voice characteristics to create a completely synthetic voice output.

The quality of a speech synthesizer is judged by its similarity to the human voice and by its ability to be understood clearly. An intelligible text-to-speech program allows people with visual impairments or reading disabilities to listen to written words on a home computer.

Long before the invention of electronic signal processing, some people tried to build machines to emulate human speech. Some early legends of the existence of speaking heads involved Gerbert of Aurillac, Albertus Magnus and Roger Bacon.

In 1779 the German-Danish scientist Christian Gottlieb Kratzenstein built models of the human vocal tract that could produce the five long vowel sounds.